

RTD PT100 to RS485 Modbus RTU Signal Converter

BRT PT100AD

Main Features

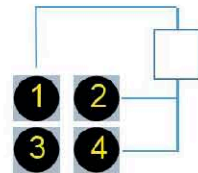
- 2-wire PT100 or 3-wire PT100 sensor signal input.
- Measuring temperature range: -199℃ --+600℃.
- Accuracy: $\pm 0.2^{\circ}\text{C}$, max. error $\pm 1^{\circ}\text{C}$.
- Standard MODBUS RTU communication protocol.
- Standard RS485 port, baud rate: 2400--115200bps adjustable.
- Max. supports 32 nodes (customized nodes available).
- Support read temperature value, modify address, baud rate, temperature correction value, etc.
- Operating voltage: DC7-30V.
- 35mm DIN Rail-mounted casing.
- Have watch dog function, automatic reset in 5 seconds in case of error.
- High accuracy high resolution AD chip used.
- Have anti-polarity reverse connection, anti-surge, over-current and EMC protection circuits.

Main Technical Parameters

Terms	Value
Input	2 -wire/3-wire PT100 sensor signal (Recommended)
Input temperature range	-199℃ --+600℃
Measuring accuracy	$\pm 0.2^{\circ}\text{C}$
Operating power supply	7-30VDC
Operating current	<50mA
Baud rate	2400--115200bps N1
Max. Nodes	32
Communication address	1-255
Operating temperature	-35℃ - +75℃
Storage temperature	-40℃ - +85℃
Size	95x25x41mm (L x W x H)

Terminals Description

Terminal No	Code	Description
1	RT+	*3-wire PT100 sensor: 2 x PT100 wires with the same color are connected to RT1- (#2) and RT2- (#4), the last wire is connected to RT1+ OR RT2+ (#1 or #3) *2-wire PT100 sensor: connect to RT1+ (#1) and RT1-(#2), short-circuited connection to RT1-(#2) and RT2- (#4)
2	RT-	
3	RT+	
4	RT-	
5	PWR+	DC6-30V operating power supply+
6	PWR-	DC6-30V operating power supply-
7	485A/D+	RS485 A
8	485B/D-	RS485 B



MODBUS Communication Protocol:

Communication Baud Rate: 2400-115200bps (default 9600),

Stop bit: 1

Data byte: 8

No checksum

Register Function

Register address	Functions	(R/W) Read/Write	Note
00	Broadcast address	R	
01	Temperature value	R	16-bit integer signed
05	Device address (default 100/0x64)	R/W	8-bit integer unsigned
06	Baud rate (default 9600)	R/W	8-bit integer unsigned
04	Correction value/modify parity bit (0, none)	R/W	8-bit integer unsigned

*Communicate with PLC, register address +40001; read channel#1 temp. value, read 40002.

Read Temperature Value (Register address is 01)

Send command: 64 03 00 01 00 01 DC 3F

Description	64	03	00	01	00	01	DC	3F
	Device address	Function code	Register starting add. higher bit	Register starting add. lower bit	Read numbers higher bit	Read numbers lower bit	CRC_H	CRC_L

Returning data: 64 03 02 01 0C F5 D9

Description	64	03	02	01	0C	F5	D9
	Device address	Function code	Data numbers	Temp. data higher bit	Temp. data lower bit	CRC_H	CRC_L

Temperature value calculation: 01 0C (signed integer: -32768~32767, Negative number is complement.)

Convert it into decimal number: 268, then divided by 10 equals 26.8, that is the current temperature value measured is 26.8℃

Modify device address (Register address is 05)

Send command: 64 06 00 05 00 01 51 FE (Modify address 100 into 1)

Description	64	06	00	05	00	01	51	FE
	Device address	Function code	Register add. higher bit	Register add. lower bit	parameters higher bit	parameters lower bit	CRC_H	CRC_L

Return data: 64 06 00 05 00 01 51 FE

Description	64	06	00	05	00	01	51	FE
	Device address	Function code	Register add. higher bit	Register add. lower bit	parameters higher bit	parameters lower bit	CRC_H	CRC_L

Note:

1. Default address is 100 that is 0x64
2. Device address setting range available: 1-247, 249 is the common address.

3. Setting device address successfully, no need to restart it, it works immediately.
4. Don't know device address, user can inquire address through broadcast address 0.

Modify Baud Rate (Register address 06)

Send command: 64 06 00 06 00 01 A1 FE (modify baud rate into 2400)

Description	64	06	00	06	00	01	A1	FE
	Device address	Function code	Register add. higher bit	Register add. lower bit	parameters higher bit	parameters lower bit	CRC_H	CRC_L

Return data: 64 06 00 06 00 01 A1 FE

Description	64	06	00	06	00	01	A1	FE
	Device address	Function code	Register add. higher bit	Register add. lower bit	parameters higher bit	parameters lower bit	CRC_H	CRC_L

Note:

1. Default baud rate is 9600.
2. Setting baud rate successfully, no need to restart it, it works immediately.
3. Register address is 03, set different parameters, the corresponding baud rate table below:

01	02	03 default	04	05	06	07
2400	4800	9600	19200	38400	57600	115200

Modify Parity Bit (Register address is 04)

Send command: 64 06 00 04 00 01 00 3E

Description	64	06	00	04	00	01	00	3E
	Device address	Function code	Register add. higher bit	Register add. lower bit	parameters higher bit	parameters lower bit	CRC_H	CRC_L

Return data: 64 06 00 04 00 01 00 3E

Description	64	06	00	04	00	01	00	3E
	Device address	Function code	Register add. higher bit	Register add. lower bit	parameters higher bit	parameters lower bit	CRC_H	CRC_L

Note:

1. Correction value range: 0-99, default 00, no parity.
2. 01 is Odd parity, 02 is even parity bit.